



Anhui MOON Technology Co., Ltd.

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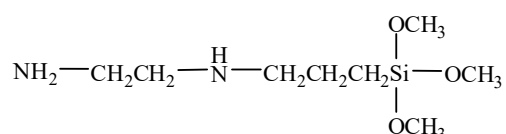
Product information

MOON-792

N-(β-Aminoethyl)-γ-aminopropyltrimethoxysilane

Product description

Structural formula:



Empirical formula: C₈H₂₂N₂O₃Si

Molecular weight: 222.4

CAS No.: 1760-24-3

Chemical name:

N-(β-Aminoethyl)-γ-aminopropyltrimethoxysilane

Properties

MOON-792 is a bifunctional silane possessing a reactive amino groups and hydrolysable methoxysilyl groups. The dual nature of its reactivity allows it to bind chemically to both inorganic materials (e.g. glass, metals, fillers) and organic polymers (e.g. thermoplastics, thermosets or elastomers). It is used as adhesion promoter at organic/inorganic interfaces, for modification of surfaces (corrosion prevention, component of primers) or silicone-polymer or as crosslinker (moisture crosslinking of polymers).

The use of MOON-792 as a coupling agent leads to in general to an improvement of mechanical and electrical product properties above all under exposure to heat and/or moisture.

MOON-792 is a colorless to slightly yellowish liquid with an amine-like odor which is soluble in alcohols and aliphatic as well as aromatic hydrocarbons.

Technical data

Typical characteristics	Value
Appearance	colorless to slightly yellowish liquid
Purity	≥97.0%
Density at 25°C	1.03g/ml
Boiling point at 760mmHg	261°C
Flash point (closed cup)	128°C
Refractive index (25°C)	1.4425

Reactivity

MOON-792 is a bifunctional organic compound in which the silicon-functional methoxy-groups hydrolyze in the presence of water to form reactive silanols, which can be bonded to an inorganic substrate; the organophilic diamino group can interact with a suitable polymer.

The hydrolysis of MOON-792 takes place autocatalytically. The pH of the hydrolysate is about 10. For longer stability of the hydrolysate a pH of 4 is recommended. Examples of particularly suitable inorganic substrates are glass or glass fibres. Examples of suitable polymers are PA, PBT, PC, EVA, modified PP, PVAC, PVC, acrylates and silicones.

MOON-792 can undergo reactions with ketones or esters as solvents. The silane or silanized substrates can react with carbon dioxide to form the corresponding carbonates. Product modifications are possible through addition reactions with suitable monomeric or polymeric compounds (such as isocyanates, epoxides).

Application and performance

MOON-792 is an important ingredient in many applications.

Examples are:

✧ glass fibre/glass fabric composites: as size



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constituent or finish

- ✧ glass and metal primers
- ✧ sealants and adhesives: as primer or additive
- ✧ mineral-filled polymers (composites): for pre-treatment of fillers and pigments or as additive
- ✧ paints and coatings: as additive and primer for improving adhesion to the substrate

The most important effects which can be achieved using MOON-792 are:

improvement in product properties, such as

- ✧ flexural strength, tensile strength, impact strength and modulus of elasticity
- ✧ moisture and corrosion resistance

improvement in processing properties, such as

- ✧ adhesion

The data presented in this leaflet are in accordance with the present state of our knowledge, but do not absolve the user from carefully checking all supplies immediately on receipt. We reserve the right to alter product constants within the scope of technical progress or new developments. The recommendations made in this leaflet should be checked by preliminary trials because of conditions during processing over which we have no control, especially where other companies' raw materials are also being used. The recommendations do not absolve the user from the obligation of investigating the possibility of infringement of third parties' rights and, if necessary, clarifying the position. Recommendations for use do not constitute a warranty, either express or implied, of the fitness or suitability of the products for a particular purpose.

Product safety, handling and storage

Customers considering the use of this product should review the latest Material Safety Data Sheet and label for product safety information, handling instructions, personal protective equipment if necessary, and any special storage condition required. The “Best use before end” date of each batch is shown on the product label. Storage beyond the date specified on the label does not necessarily

mean that the product is no longer usable. In this case however, the properties required for the intended use must be checked for quality assurance reasons.

Packaging

Information on available container sizes is obtainable from **ANHUI MOON** supplier.